

A Generative Construction of the Level-Set Family of a Square-Plate Eigenmode

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Abstract

A finite, purely combinatorial construction — sets of axes on a 12×12 square with periodic offsets, combined by a parity rule — was developed independently of any physical model, as a pattern grammar. We show that what it produces is a classical object of plate theory, reached by another route.

Precisely: twelve of the sixteen axis families are complete level sets of the single function $\varphi(x) = \cos(\pi x/3)$, taken at the seven levels $\cos \theta$ for $\theta = 0^\circ, 30^\circ, \dots, 180^\circ$; the two remaining families are the two halves of the level $\theta = 90^\circ$. The construction's involution — what its author calls *mutation* — is exactly the reflection $\theta \mapsto 180^\circ - \theta$. The construction's two reading directions, orthogonal and diagonal, are the same function read with period 12 and with period 6 — a periodicity of the function, not an identification of lines — and that difference of period accounts quantitatively for a structural asymmetry observed before it was explained: 30 shared axes, all diagonal. On the *second cut* of the diagonal vocabulary introduced in the companion paper, the diagonal side becomes as clean as the orthogonal one: each of the seven levels has its own generator, the exceptions of the drawn inventory disappear, and the mutation is $\theta \mapsto 180^\circ - \theta$ on the diagonals as well. Of the 4095 drawings of that cut, exactly $127 = 2^7 - 1$ are unions of complete level sets, and only those have a counterpart on a plate.

We then show that φ itself is an exact eigenfunction of the Kirchhoff plate with all four edges *guided* — the cylindrical mode (4,0), $k^2 = 16$ — so that the orthogonal families are the iso-amplitude lines of a genuine plate mode. The two-dimensional product $\varphi(x)\varphi(y)$ is likewise exact, of index (4,4) and $k^2 = 32$, and its nodal set is exactly the orthogonal level-0 family. For the diagonal families the situation is different and we state it separately: the degenerate combinations $\cos(4\pi x/a) \pm \cos(4\pi y/a)$ are exact guided modes of the same eigenvalue, and their nodal sets are exactly the two diagonal level-0 families — Colwell's mechanism for diagonal symmetry — while the six other diagonal levels have no guided mode behind them. The carrier (4,0) and the figure (4,4) are a factor 2 apart in frequency. One of those two nodal sets is corroborated from outside: it is the 48-cell skeleton that all sixty plates of an independent, earlier corpus carry in common, so the same family is selected once by the differential equation and once by a construction with no equation in view.

The same function is also the exact eigenfunction of a *second* physical system, and that one is trivially realisable: the acoustic pressure in a rectangular cavity with rigid walls, where the Neumann condition holds identically. There the governing equation is Helmholtz rather than biharmonic, so the frequency law is $f \propto \sqrt{m^2 + n^2}$ instead of $f \propto m^2 + n^2$. Measuring a ratio of two frequencies therefore discriminates between the two systems, which turns a reservation into a test.

Of the seven levels only one is nodal, so powder shows one of them. The other six are observable by time-averaged holographic interferometry, whose fringes are iso-amplitude contours, and we give the drive-amplitude ratios — 1:1.155:2 — that bring them successively onto the first dark fringe. That is the one falsifiable prediction here. For a *free* plate the cosine product is the classical approximation and not the exact mode. No physical measurement was performed.

1. What is being claimed

Two routes lead to the same family of curves.

The first is analytic and two centuries old: one writes the plate equation, imposes boundary conditions, and solves. The nodal lines of the solution are drawn by sand on a vibrating plate, and have been since Chladni.

The second route is the subject of this paper’s companion construction: one posits a small vocabulary of straight lines on the square, combines chosen subsets by a parity rule, and obtains figures. Nothing in that construction mentions a plate, a frequency, or a differential equation.

The claim here is that the second route reaches the level-set family of the first, exactly, and that the identification is not approximate, not statistical, and not a fit. The identification is checked line by line against the construction’s own published plates, all of which were verified region by region beforehand.

We are careful about what such a statement is worth. Arriving at a known object by an unusual road is a result about the road, not about the object. What the construction contributes is a generative grammar — with an algebra, an involution, and a closure — for a family that analysis produces only by solving.

Three kinds of statement are made, and they are kept apart throughout (§6):

status	statements
exact, checked on finite sets of lines	the identification with the level sets of φ (§3), on both cuts of the diagonals
exact in a model, not realised	the guided plate (§4.1–§4.4); “exact” for the plate always means <i>for guided edges</i> — for free edges the same statements are approximations
exhaustive on a finite corpus	the 48-cell skeleton common to the sixty published images (§4.2)
exact, and realisable	the rigid-walled cavity (§4.5)
predicted, not measured	the fringe ratio 1: 1.155: 2 (§5) and the plate/cavity frequency ratio (§4.5)

2. The construction, in brief

The construction is stated fully in the companion paper; we recall only what is needed.

The plate is the square $[0, 12]^2$ divided into 144 unit cells.

An *axis* is a line of one of four natures — horizontal, vertical, and the two diagonal directions — specified by its signed offset from the centre, measured in cells; diagonal offsets are the perpendicular coordinate rescaled to cells.

Two axes of the same nature and different offsets are **different lines**, and offsets lie in $(-6, 6]$; every count in this paper is a count of drawn lines.

One periodicity must be kept apart from another, because conflating them produces contradictions. The function φ of §3 has period 6 along a diagonal, so two diagonal offsets differing by 6 lie on the **same level** of φ — they are not the same line. The parity of §3 is taken over periodic stripe systems, and a lattice translation by (12,0) carries a diagonal line to the one 6 further along, so a single stripe system can show two of its lines inside the square and is counted once there. Seven of the sixteen families contain such pairs.

Sixteen *families* of axes are given, indexed by a level $T_0, \dots, T_3$ and by one of four names, YIN, YIN MUT, YANG, YANG MUT. The YIN families are purely orthogonal, the YANG families purely diagonal, at every level, without exception. Counted as drawn segments their union has 94 distinct axes, 30 of which lie in two families at once (§3.4); counted as band systems, the generating objects of the parity, there are 72.

A figure is obtained from a chosen union of families by a parity rule: a region is dark according to the parity of the number of axis bands separating it from a reference. The construction is reversible — a figure determines the families that produced it — and it was verified exactly, region by region, on the published corpus.

3. The identification

Let

$$\varphi(x) = \cos\left(\frac{\pi x}{3}\right).$$

3.1 The orthogonal families

Reading offsets as positions $p = 6 + e$ in $[0, 12)$, the eight orthogonal families are exactly the level sets $\varphi^{-1}(\cos \theta)$:

θ	level	positions	family
0°	1	0, 6	T_0 YIN
30°	$\frac{\sqrt{3}}{2}$	0.5, 5.5, 6.5, 11.5	T_3 YIN
60°	$\frac{1}{2}$	1, 5, 7, 11	T_2 YIN
90°	0	1.5, 4.5, 7.5, 10.5	T_1 YIN $\cup$ T_1 YIN MUT
120°	$-\frac{1}{2}$	2, 4, 8, 10	T_2 YIN MUT
150°	$-\frac{\sqrt{3}}{2}$	2.5, 3.5, 8.5, 9.5	T_3 YIN MUT
180°	-1	3, 9	T_0 YIN MUT

Six of these seven are complete families. The level $\theta = 90^\circ$ is the one exception: the construction splits it into two families, T_1 YIN (1.5, 10.5) and T_1 YIN MUT (4.5, 7.5), whose union is the level.

3.2 The mutation is a reflection of the angle

Proposition 1. For each level $t \in \{0, 2, 3\}$, if T_t YIN is the level set at angle θ , then T_t YIN MUT is the level set at $180^\circ - \theta$.

The map $\theta \mapsto 180^\circ - \theta$ is $\varphi \mapsto -\varphi$, i.e. the exchange of the two phases of the standing wave. The level T_1 is its own mutant because 90° is the fixed point of the reflection, and it is there, and only there, that base and mutant coincide — which is why the construction lists them as two families with a single level set.

3.3 The diagonal families: the same function, half the period

Let $v = (x + y)/2$ be the diagonal coordinate in cells. A lattice translation by (12,0) shifts v by 6, so along a diagonal the function is periodic of period 6, not 12. Reducing modulo 6:

family	=	levels
T_0 YANG	T_0 YIN	0°
T_0 YANG MUT	T_0 YIN MUT	180°
T_1 YANG	T_0 YIN $\cup$ T_0 YIN MUT	$0^\circ, 180^\circ$
T_1 YANG MUT	T_1 YIN $\cup$ T_1 YIN MUT	90°
T_2 YANG	T_2 YIN $\cup$ T_2 YIN MUT	$60^\circ, 120^\circ$
T_2 YANG MUT	T_3 YIN $\cup$ T_3 YIN MUT	$30^\circ, 150^\circ$
T_3 YANG	T_3 YIN $\cup$ T_3 YIN MUT	$30^\circ, 150^\circ$
T_3 YANG MUT	T_2 YIN $\cup$ T_2 YIN MUT	$60^\circ, 120^\circ$

Theorem 1. Every one of the sixteen families is contained in a union of level sets of φ at angles that are multiples of 30° . Twelve are complete — a whole level, or the whole union of two. The four exceptions are:

- T_1 YIN and T_1 YIN MUT, the two halves of the level 90° : positions $\{4.5, 7.5\}$ and $\{1.5, 10.5\}$;
- T_2 YANG, the level 120° together with the level 60° less its two corner lines, offsets ± 5 ;
- T_3 YANG, the level 150° together with the level 30° less its two corner lines, offsets ± 5.5 .

The four missing corner lines are carried by T_3 YANG MUT and T_2 YANG MUT respectively, and that is why those pairs share axes.

The last two exceptions are exceptions of the *inventory of drawn segments*, not of the identification. A diagonal offset and its translate by 6 define the same band system, so as band systems T_2 YANG and T_3 YANG MUT are one and the same object, as are T_3 YANG and T_2 YANG MUT; the two corner lines that the inventory assigns to one member of each pair are already carried by the other. Read on figures — which is what the parity rule produces and what the plates show — the identification with the level sets is therefore exact for all sixteen families, and the pair of exceptions at 90° is the only one that survives. The companion paper on the construction records the same fact as two of the three relations of its code.

3.4 Two consequences that were observed before they were explained

The shared axes. Of the 94 axes, 30 belong to two families each, and all 30 are diagonal; no orthogonal axis is shared. The table above gives the reason: T_2 YANG and T_3 YANG MUT are the same pair of levels, as are T_2 YANG MUT and T_3 YANG. On the orthogonal side, period 12 keeps the seven levels disjoint; on the diagonal side, period 6 folds them onto each other.

The asymmetry of the mutation. The construction’s rule “offset $+3 \pmod{6}$ ” relates base to mutant on all four orthogonal levels and on none of the diagonal ones. The table shows why: each diagonal family already contains both θ and $180^\circ - \theta$, so base and mutant are not separated there and no shift can relate them.

Neither fact was used to obtain the identification; both follow from it. And both are facts of the *first cut* of the diagonals, not of the diagonals themselves, as the next section shows.

3.5 The second cut: one level per generator

The companion paper (§5.5 there) examines a second cut of the diagonal vocabulary, drawn after the first and after the identification above, so that it is a cleaner vocabulary and not further independent evidence: twelve elements, YA1–YA6 and AY1–AY6, made of unit segments along cell diagonals, pairwise disjoint and together exactly the 288 cell diagonals of the square. Read against φ , with $c = 2e$ the signed offset of a diagonal line ($x + y = 12 + c$ or $y - x = c$) and its level $\cos(\pi c/6)$:

θ	level	second cut	first cut
0°	1	YA6	T_0 YANG
30°	$\frac{\sqrt{3}}{2}$	YA5 $\cup$ AY1	T_2 YANG MUT, T_3 YANG (with 150°)
60°	$\frac{1}{2}$	YA4 $\cup$ AY2	T_2 YANG, T_3 YANG MUT (with 120°)
90°	0	YA3 $\cup$ AY3	T_1 YANG MUT
120°	$-\frac{1}{2}$	YA2 $\cup$ AY4	T_2 YANG, T_3 YANG MUT (with 60°)
150°	$-\frac{\sqrt{3}}{2}$	YA1 $\cup$ AY5	T_2 YANG MUT, T_3 YANG (with 30°)
180°	−1	AY6	T_0 YANG MUT

Every element lies on a single level — YAk on $180^\circ - 30k^\circ$, AYk on $30k^\circ$ — and the elements of a level tile its lines exactly, checked segment by segment. As band systems $YA(6-j)$ and AYj coincide, so the second cut gives seven generators, one per level, against five level sets for the eight diagonal families of the first cut. Three consequences, each an exact check:

- **No exceptions.** Every generator of the second cut is a complete level set. The inventory exceptions of Theorem 1 — corner lines carried by the wrong member of a pair — have no analogue, and the second cut has no shared axis at all.
- **The mutation is the reflection on the diagonals too.** The lines of AYk are those of YAk shifted by 3 in offset, and YAk, AYk sit at θ and $180^\circ - \theta$. The asymmetry of §3.4 was a property of the first cut.
- **Which drawings a plate can show.** Of the 4095 non-empty unions of the twelve elements, exactly $127 = 2^7 - 1$ are unions of complete level sets — the two elements of a level taken together or not at all. A level set of φ is a set of whole lines, so these 127 drawings, and no others, have a counterpart among the iso-amplitude figures of §4 and §5.

On the second cut the diagonal reading is thus as complete as the orthogonal one: seven levels, seven generators, and the involution $\theta \mapsto 180^\circ - \theta$ exchanging the yang element with its mutable twin.

4. The plate

4.1 The orthogonal carrier is a one-dimensional mode

The orthogonal families of §3 are level sets of the one-variable function φ , and this must be said precisely, because the distinction decides what is measurable.

Let $W(x, y) = \cos(m\pi x/a)$, constant in y . Symbolic evaluation gives

$$\nabla^4 W = \frac{\pi^4 m^4}{a^4} W, \quad \frac{\partial W}{\partial n} = 0 \quad \text{and} \quad V_n = 0 \quad \text{on all four edges,}$$

which are the conditions of a **guided** (also *sliding*) edge.

Theorem 2. $\varphi(x) = \cos(4\pi x/12)$ is an exact eigenfunction of the square Kirchhoff plate of side 12 with all four edges guided: the mode (4,0), cylindrical bending, with $k^2 = m^2 + n^2 = 16$.

Its level sets are straight lines parallel to the edges, and they are the eight orthogonal families of §3, taken in x and in y .

4.2 The diagonal families are not level sets of a guided mode

It would be convenient to write the diagonal families as level sets of $\cos(2\pi(x+y)/a)$. That function is not a mode of the guided plate: symbolic evaluation gives, at $x = 0$,

$$\frac{\partial W}{\partial n} = -\frac{2\pi}{a} \sin \frac{2\pi y}{a} \neq 0,$$

so the guided condition fails on every edge. It is a mode of the torus, not of the plate, and we do not use it.

What is true is stronger, and it is Colwell's mechanism [5]. The modes (4,0) and (0,4) are degenerate — both have $k^2 = 16$ — and their combinations

$$\cos \frac{4\pi x}{a} \pm \cos \frac{4\pi y}{a}$$

are exact guided modes of that same eigenvalue $256\pi^4/a^4$, verified symbolically on all four edges.

Theorem 3. The nodal set of $\cos(4\pi x/a) + \cos(4\pi y/a)$ is exactly the family T_1 YANG MUT (diagonal offsets $\pm 1.5, \pm 4.5$ in both senses), and the nodal set of the difference is exactly T_1 YANG (offsets 0, ± 3).

The mechanism is the one Colwell exploited [5], stated in its general form: for any pair (m,n), the difference $\cos \frac{m\pi x}{a} \cos \frac{n\pi y}{a} - \cos \frac{n\pi x}{a} \cos \frac{m\pi y}{a}$ vanishes on $x = y$, and the sum vanishes on $x + y = a$ according to the parity of $m + n$. Colwell's own statement concerns pairs of *different* parity, where the principal diagonals are nodal; Theorem 3 is the same mechanism applied to the pair (4,0), (0,4), which have the same parity and give not the principal diagonals but the parallels at offsets $\pm 1.5, \pm 4.5$ and 0, ± 3 . We say this so that a reader does not look in Colwell for a sentence that is not there.

The six remaining diagonal levels are iso-amplitude lines of a one-variable function in $x + y$, and **no guided plate mode stands behind them**. We state this rather than paper over it: the correspondence is complete on the orthogonal side and partial on the diagonal one.

A corroboration that comes from outside this paper. Theorem 3 singles out one family, T_1 YANG, as the nodal set of $\cos(4\pi x/a) - \cos(4\pi y/a)$. That family was not selected here for its physical status; it was selected by the equation. It turns out to be the one family that a *different* corpus — sixty plates built years earlier under a separate set of rules, and described in the companion paper on the construction — carries in common. In each of the sixty, the 48 cells whose centres lie on the axes of T_1 YANG form one monochrome region, identically in all sixty; the remaining 96 cells carry the parity of a chord of the four T_0 families, which is what varies from plate to plate. The skeleton those sixty images share is therefore exactly the nodal set of a degenerate guided mode of eigenvalue $256\pi^4/a^4$. We did not arrange this: the selection was made once by the differential equation and once by an author who had no equation in view, and the two selections agree. It is the strongest corroboration in the paper that does not require a measurement.

4.3 Only the zero level belongs to the two-dimensional mode

The product $\varphi(x)\varphi(y) = \cos(4\pi x/12)\cos(4\pi y/12)$ is likewise an exact guided mode, of index (4,4) and $k^2 = 32$, with the same eigenvalue formula $\pi^4(m^2 + n^2)^2/a^4$.

Its level sets, however, are **not** straight except at zero. Along the line $x = 1$ of the family T_2 YIN, where $\varphi(x) = \frac{1}{2}$, the product $\varphi(x)\varphi(y)$ runs from $+\frac{1}{2}$ to $-\frac{1}{2}$ as y varies. Only at $\varphi = 0$ does a straight line lie wholly in a level set of the product, and there it lies in the zero set.

Corollary. *The orthogonal level-0 family, T_1 YIN $\cup$ T_1 YIN MUT, is exactly the nodal set of the guided mode (4,4). The diagonal level-0 families are not: they are the nodal sets of the degenerate combinations of §4.2, by Theorem 3. The remaining six levels are iso-amplitude lines of the one-dimensional mode (4,0), and are not level sets of any two-dimensional product mode.*

We state this because the looser formulation — “the families are level curves of a square-plate mode” — is false for six of the seven levels, and a reader would find it out.

4.4 Spectrum, and a convention on k^2

For the square, the cosine and the sine families share the eigenvalue $\pi^4(m^2 + n^2)^2/a^4$, so the guided plate and the simply supported plate have the same spectrum and differ only in eigenfunction. Hence

$$f = Ck^2, \quad k^2 = m^2 + n^2, \quad C = \frac{\pi}{2} \sqrt{\frac{D}{\rho h}} \frac{1}{a^2}.$$

The constant C is a calibration and not a result: it fixes the plate, not the geometry.

Two conventions for k^2 are in circulation in the associated material and must not be confused. Counting half-wavelengths across the side a — the standard plate convention, used here — gives (4,4) and $k^2 = 32$. Counting whole periods across the 12-cell grid, as a discrete Fourier transform of the figure does, gives (2,2) and $k^2 = 8$. The two are related by

$$k_{\text{grid}}^2 = \frac{1}{4}(m^2 + n^2),$$

so a calibration written $f = 128 k_{\text{grid}}^2$ is the same law as $f = 32(m^2 + n^2)$.

The two modes of §4.1 and §4.3 are not at the same frequency, and it matters which one a stated figure refers to. The **carrier** (4,0) has $k^2 = 16$, $k_{\text{grid}}^2 = 4$, hence $f = 512$ Hz at this calibration. The **figure** (4,4) — the parity of the orthogonal families together, and the pattern powder would draw — has $k^2 = 32$, $k_{\text{grid}}^2 = 8$, hence $f = 1024$ Hz. The ratio is exactly 2, and the fringe experiment of §5 is performed on the carrier, therefore at half the frequency of the sand figure.

4.5 The same function, a second equation, a realisable system

The guided plate is an idealisation. The function is not.

Let $W(x, y) = \cos(m\pi x/a) \cos(n\pi y/b)$ and let it now be a pressure field rather than a displacement. Symbolic evaluation gives

$$\nabla^2 W = -\pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right) W, \quad \frac{\partial W}{\partial n} = 0 \text{ on all four walls.}$$

That is the Helmholtz equation with the Neumann condition, which is exactly the acoustic problem for a rectangular cavity with rigid walls. The condition is satisfied identically, not approximately, and such a cavity is any rectangular box. One qualification: the modes in play are those of vertical index zero, the pressure being constant over the height, so the figure is read on a horizontal section and the box must be driven below the first vertical cut-off, or be shallow enough that it is. With a non-zero vertical index the field varies through the height and the horizontal figure is unchanged but its amplitude is not.

Theorem 4. *On the rectangle $a \times b$, the function $\cos(m\pi x/a) \cos(n\pi y/b)$ is simultaneously an exact eigenfunction of two different problems: the biharmonic problem with guided edges, with eigenvalue*

$$\pi^4 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right)^2,$$

and the Helmholtz problem with rigid walls, with eigenvalue

$$\pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right).$$

The same is true of the one-dimensional carrier. On the square $a = b$ these read $\pi^4 (m^2 + n^2)^2 / a^4$ and $\pi^2 (m^2 + n^2) / a^2$, which is the case $\varphi(x) \varphi(y)$ of §4.3.

The two problems share their eigenfunctions and not their spectra; on the square,

$$\text{plate: } f \propto m^2 + n^2, \quad \text{cavity: } f \propto \sqrt{m^2 + n^2}.$$

This is the useful part. A ratio of two measured frequencies separates the two laws immediately — for the carrier and the figure of §4.4, the plate gives $f_{4,4}/f_{4,0} = 2$ and the cavity gives $\sqrt{2}$. And for the grid values $k^2 = 1, 4, 9$ the plate law gives frequencies in the ratio 1:4:9 while the cavity law gives 1:2:3, the harmonic series.

A third realisation has the same separable form under linear theory: the free surface of a liquid in a rectangular basin with vertical walls, where the elevation of a sloshing mode is $\cos(m\pi x/a) \cos(n\pi y/b)$ and the seven levels are iso-elevation lines. Its frequency law is neither of the two above: sloshing is dispersive, $\omega^2 = gk \tanh kh$ with k the horizontal wavenumber and h the depth, so what the three systems share is the *figure* and not the spectrum, and the discriminating ratio of the previous paragraph does not transfer to the tank. Excited in a one-dimensional mode, those lines are straight, and they are visible under grazing light. This is the least expensive experiment in the whole programme: a rectangular tank, a loudspeaker beneath it, a lamp.

5. What is nodal, what is not, and what can be seen

Of the seven levels, exactly one is a zero set: $\theta = 90^\circ$. The other six are iso-amplitude lines. Powder on a plate accumulates only on the first, so the six others are invisible to Chladni's method.

They are not invisible to optics. In time-averaged holographic interferometry (Powell and Stetson, 1965) and in its digital successor, electronic speckle pattern interferometry, the reconstructed intensity of a harmonically vibrating surface is modulated by J_0^2 of the local amplitude, so the dark fringes lie at the **zeros of J_0** and each fringe is a contour of constant vibration amplitude. The six remaining levels are therefore observable by a method established for sixty years.

The correspondence is not automatic, and we state the experiment rather than gesture at it. The fringes fall at amplitudes w satisfying $4\pi w/\lambda = j_{0,k}$, where $j_{0,1} = 2.405$, $j_{0,2} = 5.520, \dots$ — values fixed by optics, not by the mode. To bring a chosen level $\cos \theta$ onto the first dark fringe one sets the drive amplitude A so that $A|\cos \theta| = j_{0,1}\lambda/4\pi$. For the three non-zero level pairs this is a ratio of drive amplitudes

$$\cos \theta = 1 : \frac{\sqrt{3}}{2} : \frac{1}{2} \quad \Rightarrow \quad A \propto 1 : 1.155 : 2.$$

On the one-dimensional mode (4,0) the fringes are straight, and this is the configuration in which the prediction is sharp: **three drive amplitudes in the ratio 1:1.155:2 should bring the families T_0 , T_3 and T_2 successively onto the first dark fringe, at the positions tabulated in §3.1.** On the two-dimensional mode (4,4) the fringes are the curves $\varphi(x)\varphi(y) = \text{const}$, which are not straight, and the prediction does not apply — by §4.3.

Two precisions complete the protocol. The optical constant is $\Omega \approx (4\pi/\lambda)w$ at normal illumination and observation, and it is the only one. And since the intensity goes as J_0^2 , the fringe pattern does not see the sign of the amplitude: each drive level shows the pair $(\theta, 180^\circ - \theta)$ at once, so a family and its mutant appear together — the involution of §3.2 is directly visible in a single exposure. At $\cos \theta = \pm 1$ the fringe falls on the lines of maximum amplitude, while the nodal line remains the bright fringe of order zero throughout.

One design consequence follows from §4.2 and improves the protocol. On a square plate the modes (4,0) and (0,4) are degenerate, so a point excitation generally produces a combination of the two, whose fringes are curved — which is exactly what Theorem 3 describes. For the ratio 1:1.155:2 to be clean, the degeneracy must be lifted: either by driving along a line parallel to one edge rather than at a point, or by using a **rectangular** plate $a \times b$ with $b \neq a$. Since $\cos(4\pi x/a)$ does not depend on b , the prediction carries over to the rectangle unchanged, and the square is not required. The experiment is therefore easier than the identification that motivates it.

This converts a reservation into a falsifiable statement, and it is the only one in this paper that a laboratory can refute.

6. What is established, what is measurable, and what is not

It is worth separating three kinds of statement, because they do not carry the same weight.

Verified on the data, and needing no measurement. The containment of the sixteen families in the level sets of φ (§3), complete for twelve of them and explicit about the four exceptions; on the second cut of the diagonals, one complete level per generator, the mutation as reflection, and the 127 drawings that are unions of whole level sets (§3.5). It is an identity between two finite sets of lines, checked exactly. To it we add the coincidence of §4.2: the family the equation selects, T_1 YANG, is the 48-cell skeleton common to all sixty plates of the independent corpus — again an exact check on a finite set, not an estimate.

A theorem, and likely to remain one. The guided plate (§4.1, §4.2). No plate with guided edges appears to have been realised; the condition is a mathematical idealisation, and we do not present it as a physical claim.

Measurable, and not yet measured. Four things, and the first is new.

The discriminating frequency ratio of §4.5. The two equations share the eigenfunction and differ in the spectrum, so a single ratio of measured frequencies says which law governs a given realisation: 2 against $\sqrt{2}$ for the carrier and the figure, 1:4:9 against 1:2:3 for the grid values $k^2 = 1, 4, 9$. This is cheap, and it is the first measurement we would make.

Then three more. First, the cosine product as an approximation for the *free* square plate — Chladni's own object — where the exact modes are products of free-free beam functions, combinations of $\cosh$ and $\cos$; the cosine form was introduced for square plates by Colwell in 1932 [2], and Ritz's 1909 treatment, reproduced by Gander and Wanner [10], remains the oldest quantitative agreement with measurement on the free square plate. Second, the law $f \propto m^2 + n^2$, exact for the simply supported and the guided plate: the sharp test is the ratio $f_{4,4}/f_{2,2}$, which is exactly 4 for both, and is not 4 for a free plate. Third, the fringe experiment of §5.

For calibration of what counts as agreement: a recent free-plate study reports a maximum deviation of 6% between measurement and computation [11], so a discrepancy below that threshold distinguishes nothing. A free acrylic square

of side 24 cm gives measured frequencies 107, 283, 511, 744, 1181 Hz [12], whose ratios 2.64, 4.78, 6.95, 11.0 are not those of $m^2 + n^2$ — as expected for a free plate, and a useful reminder of the size of the effect.

Not done. No plate was excited, no powder observed, no frequency measured, no fringe recorded. Everything here is computation on published drawings and symbolic verification of a differential equation.

Crossings. The figures are unions of straight lines and therefore have crossings. In the zero set of a generic real function crossings do not occur; they require a symmetry. A point-driven plate away from resonance responds as a weighted superposition of modes, whose nodal lines avoid crossing, while at an isolated resonance a single mode dominates and the straight-line grid returns [3,4]. The family described here is the symmetry-determined subclass, and we claim nothing about the generic case.

7. Prior work

The equation of a square Chladni plate in the cosine form, $A \cos \frac{m\pi x}{a} \cos \frac{n\pi y}{a} + B \cos \frac{n\pi x}{a} \cos \frac{m\pi y}{a} = 0$, is Colwell’s [2]; his companion paper on diagonal symmetry shows that when m is even and n odd, or the converse, one diagonal of the square is a nodal line [5], which is the diagonal family of the present construction, named and dated. Colwell and Hill later found that patterns produced by electrical excitation and by mechanical impact were “exactly similar” [6]. Waller’s studies of compounded modes on free square plates remain the reference for the superposition regime [7]. Warburton’s fifteen cases for the rectangular plate [8] and Leissa’s catalogue [9] situate the boundary conditions; Tuan and co-workers give the modern point-driven account [3,4].

8. Reproducibility

Every table in this paper is printed by a script that reads the published drawings and reruns in minutes; the table and the counts of §3.5 are printed by `niveaux_v2.py`, from `axes-v2.json`, in the deposit of the companion paper. Axis offsets are extracted from the vector sources and compared to level sets by exact arithmetic, not by numerical fitting. The plate identity of §4.1 is verified symbolically. The underlying plates were themselves checked region by region — 1152 regions each — before any identification was attempted, and two corrupt files found in that check were corrected before use.

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